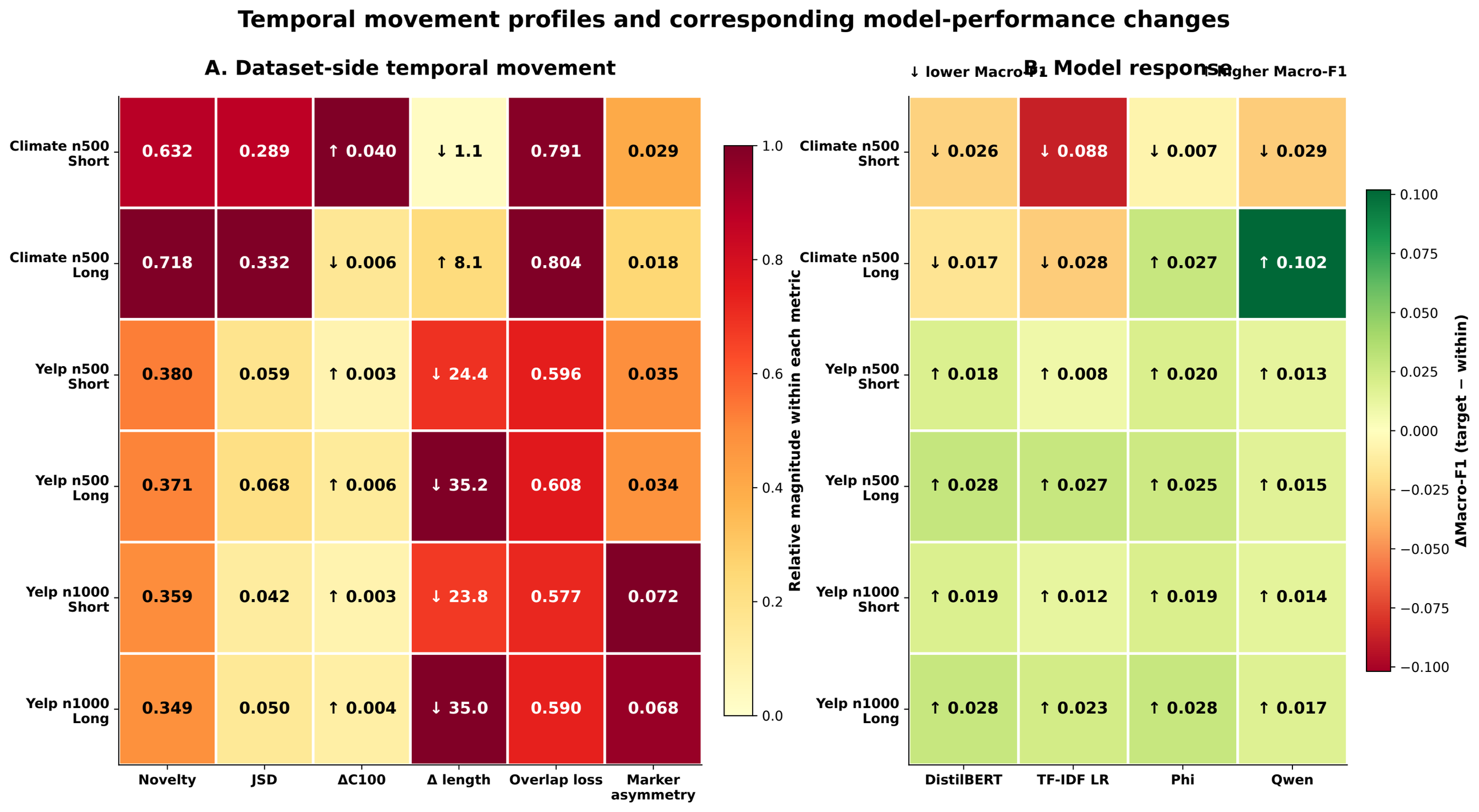
Temporal movement profiles and corresponding model-performance changes
A. Dataset-side temporal movement
B. Model response
↓ lower Macro-F1
↑ higher Macro-F1
Climate n500 Short
Climate n500 Long
Yelp n500 Short
Yelp n500 Long
Yelp n1000 Short
Yelp n1000 Long
0.632
0.289
↑ 0.040
↓ 1.1
0.791
0.029
0.718
0.332
↓ 0.006
↑ 8.1
0.804
0.018
0.380
0.059
↑ 0.003
↓ 24.4
0.596
0.035
0.371
0.068
↑ 0.006
↓ 35.2
0.608
0.034
0.359
0.042
↑ 0.003
↓ 23.8
0.577
0.072
0.349
0.050
↑ 0.004
↓ 35.0
0.590
0.068
Novelty
JSD
ΔC100
Δ length
Overlap loss
Marker asymmetry
Relative magnitude within each metric
1.0
0.8
0.6
0.4
0.2
0.0
↓ 0.026
↓ 0.088
↓ 0.007
↓ 0.029
↓ 0.017
↓ 0.028
↑ 0.027
↑ 0.102
↑ 0.018
↑ 0.008
↑ 0.020
↑ 0.013
↑ 0.028
↑ 0.027
↑ 0.025
↑ 0.015
↑ 0.019
↑ 0.012
↑ 0.019
↑ 0.014
↑ 0.028
↑ 0.023
↑ 0.028
↑ 0.017
DistilBERT
TF-IDF LR
Phi
Qwen
ΔMacro-F1 (target − within)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100